Data Quality Report — Cars Dataset ETL

Author: Mai Kie **Date:** 2026-11-03 **Source:** `cars_data.csv` (1218 rows) **Target:** `dw.fact_cars`

1. Star Schema Diagram

```
                    +------------------+
                    |   dim_company    |
                    |------------------|
                    | company_key (PK) |
                    | company_name     |
                    +--------+---------+
                             |
                             | 1 : N
                             |
+-------------------+      +--v---------------+
|   fact_cars       |      |     dim_car      |
|-------------------|      |------------------|
| car_key     (FK) +----->| car_key     (PK) |
| horsepower        |      | car_name         |
| top_speed_kmh     |      | company_key (FK) |
| acceleration_sec  |      | fuel_type        |
| price_usd         |      | seats            |
+-------------------+      | engine           |
                           +------------------+

                           Grain of fact_cars: one row per specific car
model/trim level.

2. Issues Found in Source CSV
#   Issue   Details
1   Unicode Encoding Error  File contained special characters (byte 0x96)
that crashed standard UTF-8 reading.
2   Dirty Currency Values   Prices contained dollar signs and commas (e.g.,
"$1,100,000 ").
3   Price Ranges    Some prices were ranges instead of single numbers
(e.g., "$12,000-$15,000").
4   Text in Numeric Fields  Horsepower, Speed, and Acceleration contained
text suffixes (e.g., "963 hp", "340 km/h").
5   Double Decimals Malformed numbers with multiple dots (e.g., "7.25.4").
6   Missing Data    1 row had missing critical information (Company or
Price).
3. Rules Applied
Encoding Fix: Read the file using latin1 encoding in Python
(pd.read_csv(..., encoding='latin1')).

Price Cleaning: Used Python Regex and SQL SPLIT_PART to remove $, ,, and
split ranges at the hyphen - to keep only the first number.

Numeric Extraction: Used PostgreSQL REGEXP_REPLACE(column, '[^0-9.]', '',
```

```
'g') to strip all letters and symbols from Horsepower, Speed, and
Acceleration.

Decimal Fix: Wrapped numeric extraction in SUBSTRING to prevent multiple
decimals from breaking the NUMERIC cast.

Null Guard: Dropped rows where Company Names or Cars Prices were
missing/empty.

Deduplication: Used ON CONFLICT (car_name) DO NOTHING when loading dim_car
to prevent duplicates.

4. Rows Rejected
Reason  Count
Missing critical data (Price/Company)   1
Total rejected  1
Rows loaded into fact_cars  1217
5. Data Insights (Top 5 Most Expensive Brands)
Bugatti (Avg: ~$5.9M)

Aston Martin (Avg: ~$752K)

Lamborghini (Avg: ~$650K)
```